

Project 1: Railway Gauges Data Analysis

Analytics Task Report — Day-17

1. Objective

This report documents the end-to-end analysis of the Indian Railway Gauges dataset (railway_gauges.csv), covering data loading, cleaning, visualization, filtering, feature engineering, and trend analysis across five scenarios of increasing complexity, as specified in the Day-17 Analytics Task brief.

2. Dataset Overview

The dataset contains yearly records from 1964-65 to 2012-13 (49 rows), with the following columns:

- Year — financial year range (e.g. 1964-65)
- Broad Gauge — number of broad gauge tracks
- Metre Gauge — number of metre gauge tracks
- Narrow Gauge — number of narrow gauge tracks
- Total — sum of all gauge tracks for that year

No missing values were found in any column during the cleaning step.

3. Scenario 1 — Data Loading & Cleaning

The CSV was loaded into a Pandas DataFrame. The first 5 rows and column names were displayed, missing values were checked (none found), and all gauge columns were confirmed / converted to numeric types.

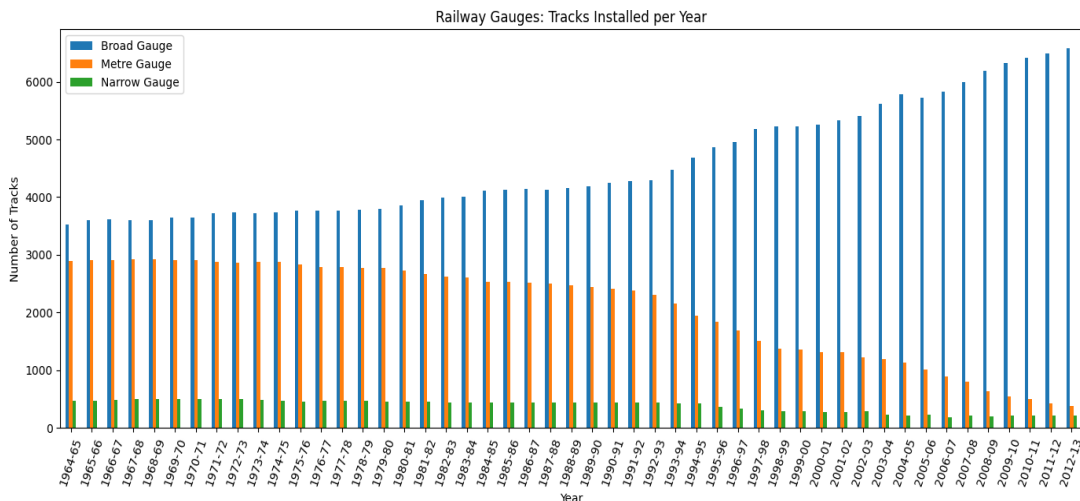


Figure 1: Broad, Metre, and Narrow Gauge tracks by year

4. Scenario 2 — Simple Visualization

The Year and Total columns were extracted and plotted as a line graph to observe the overall growth trend.

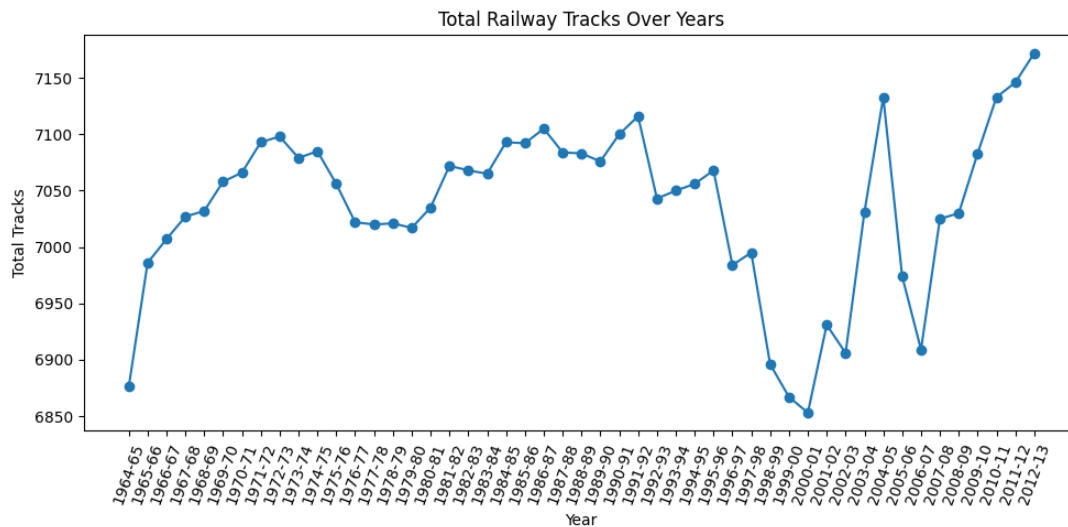


Figure 2: Total railway tracks over years

Insight: The total number of tracks rose from 6,876 (1964-65) to 7,172 (2012-13). The trend is broadly increasing, with a visible dip in the late 1990s to early 2000s before recovering and climbing further.

5. Scenario 3 — Filtering + Bar Chart

Data was filtered for years after 2000, and a grouped bar chart was plotted comparing Broad, Metre, and Narrow Gauge tracks.

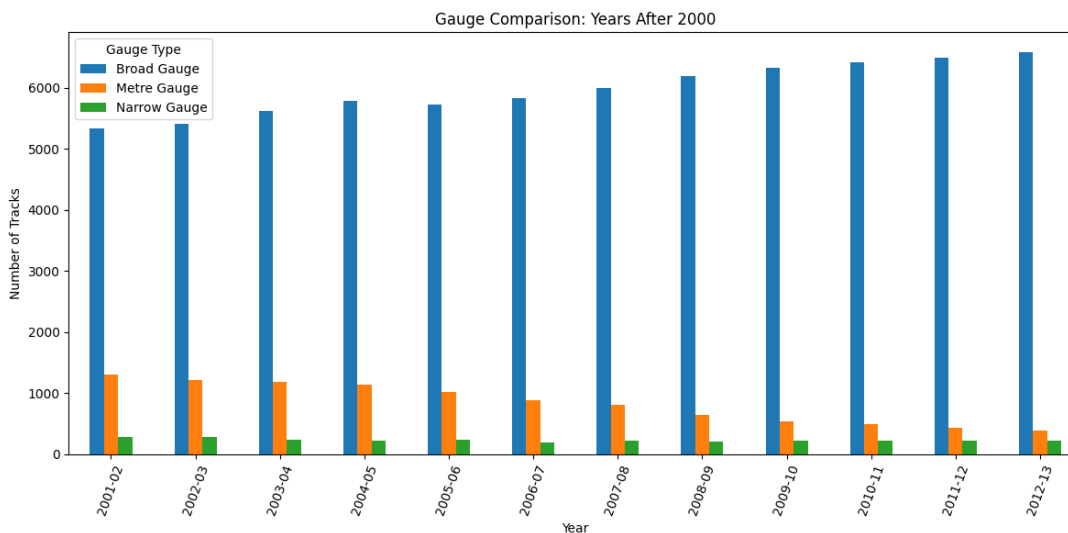


Figure 3: Gauge comparison for years after 2000

Insight: Broad Gauge dominates in recent years, growing steadily each year, while Metre Gauge is shrinking rapidly and Narrow Gauge remains marginal.

6. Scenario 4 — Feature Engineering + Pie Chart

The total contribution of each gauge type was calculated across all years and visualized as a pie chart with percentage labels.

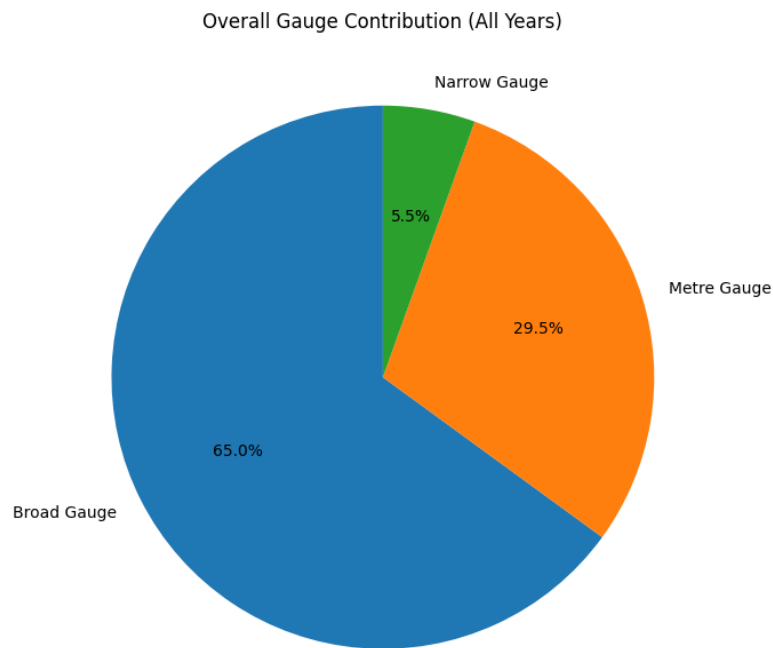


Figure 4: Overall gauge contribution (1964-2013)

Insight: Broad Gauge accounts for 65.0% of all tracks laid across the full period, followed by Metre Gauge at 29.5% and Narrow Gauge at 5.5%.

7. Scenario 5 — Advanced Analysis

Percentage columns were engineered for each gauge type, and NumPy's `diff()` function was used to calculate year-over-year growth in Total tracks. Two further visualizations were produced: a combined line graph of all three gauges, and a stacked bar chart showing composition over time.

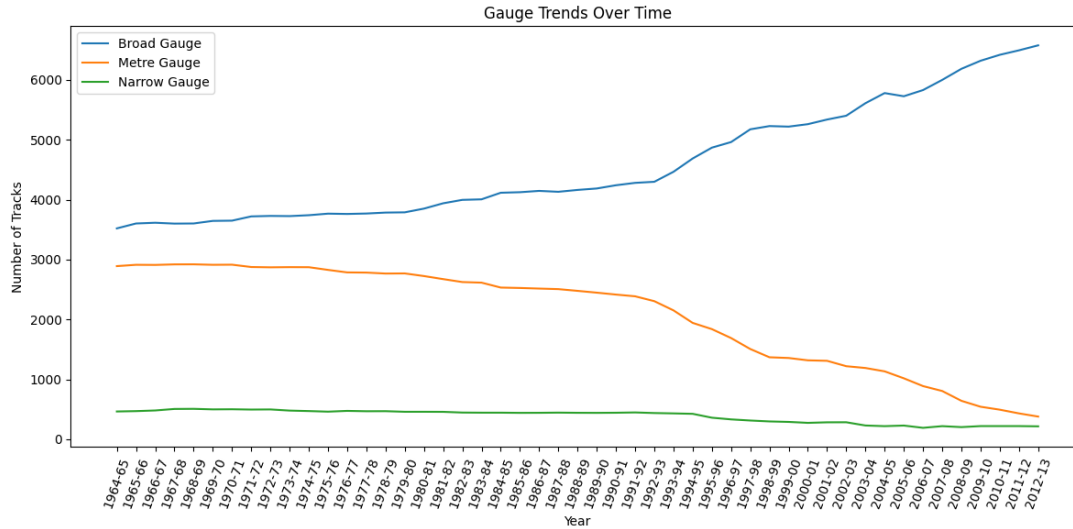


Figure 5: Trend of all three gauge types over time

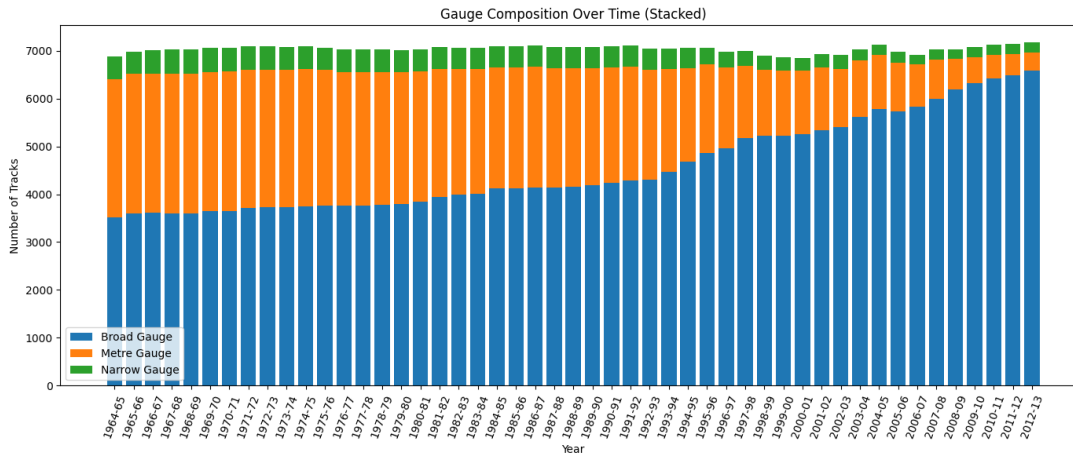


Figure 6: Gauge composition over time (stacked)

Key Findings

- Highest year-over-year growth: 2003-04 (+125 tracks)
- Largest year-over-year decline: 2005-06 (-159 tracks)
- Metre Gauge fell from 2,891 to 379 tracks over the period — a steep and sustained decline
- Narrow Gauge fell from 464 to 216 tracks — a smaller but steady decline
- Broad Gauge's share of the network rose from 51.2% to 91.7% between 1964-65 and 2012-13

8. Final Conclusion

Yes — the Indian railway system is clearly shifting toward a single dominant gauge. Broad Gauge's share of the total network rose from roughly half to over 90% across the analysis period, while both Metre and Narrow Gauge steadily lost track share. This reflects the real-world Project Unigauge initiative, under which India progressively converted its Metre and Narrow Gauge lines to Broad Gauge for standardization and operational efficiency.

9. Tools & Techniques Used

- Pandas — data loading, cleaning, filtering, grouping, feature engineering
- NumPy — array conversion, diff() for growth calculation
- Matplotlib — line chart, grouped bar chart, pie chart, stacked bar chart